

3 章 関数とグラフ
2 いろいろな関数
BASIC

■171 (1)

$$\begin{aligned}f(-x) &= 3(-x)^2 \\&= 3x^2 = f(x) \\&\therefore \text{偶関数}\end{aligned}$$

■171 (2)

$$\begin{aligned}f(-x) &= -x - 1 \\&\therefore \text{いずれでもない}\end{aligned}$$

■171 (3)

$$\begin{aligned}f(-x) &= -2 \cdot (-x) \\&= 2x = -f(x) \\&\therefore \text{奇関数}\end{aligned}$$

■171 (4)

$$\begin{aligned}f(-x) &= (-x)^2 + (-x) \\&= x^2 - x \\&\therefore \text{いずれでもない}\end{aligned}$$

■171 (5)

$$\begin{aligned}f(-x) &= (-x)^4 + 5(-x)^2 - 3 \\&= x^4 + 5x^2 - 3 = f(x) \\&\therefore \text{偶関数}\end{aligned}$$

■171 (6)

$$\begin{aligned}f(-x) &= (-x - 1)^5 \\&= -(x + 1)^5 \\&\therefore \text{いずれでもない}\end{aligned}$$

■171 (7)

$$\begin{aligned}f(-x) &= \frac{1}{4}(-x)^3 \\&= -\frac{1}{4}x^3 = -f(x) \\&\therefore \text{奇関数}\end{aligned}$$

■171 (8)

$$\begin{aligned}f(-x) &= \frac{1}{(-x)^2} \\&= \frac{1}{x^2} = f(x) \\&\therefore \text{偶関数}\end{aligned}$$

■171 (9)

$$\begin{aligned}f(-x) &= \frac{\{(-x) + 3\}^2}{-x} \\&= -\frac{(x + 3)^2}{x} = -f(x) \\&\therefore \text{奇関数}\end{aligned}$$

■172 (1)

$$\begin{aligned}y &= (x - 1)^3 + 2 \\y &= x^3 \text{ のグラフを } x \text{ 軸方向に } 1、 \\&\quad y \text{ 軸方向に } 2 \text{ 平行移動したもの}\end{aligned}$$

■172 (2)

$$y = -(x + 1)^3 + 2$$

■172 (3)

$$y = (x + 1)^4 - 2$$

■172 (4)

$$y = -(x - 2)^4 + 1$$

■173 (1)

$$\begin{aligned}y &= \frac{1}{x - 2} + 1 \quad (x \neq 2) \\&\text{漸近線 } x = 2, y = 1\end{aligned}$$

■173 (2)

$$\begin{aligned}y &= \frac{2}{x + 1} - 2 \quad (x \neq -1) \\&\text{漸近線 } x = -1, y = -2\end{aligned}$$

■173 (3)

$$\begin{aligned}y &= -\frac{1}{x - 3} + 1 \quad (x \neq 3) \\&\text{漸近線 } x = 3, y = 1\end{aligned}$$

■174 (1)

$$y = \frac{x+3}{x+1}$$

$$= 1 + \frac{2}{x+1} \quad (x \neq -1)$$

漸近線 $x = -1, y = 1$

■174 (2)

$$y = \frac{2x-5}{x-2}$$

$$= 2 - \frac{1}{x-2} \quad (x \neq 2)$$

漸近線 $x = 2, y = 2$

■175 (1)

$$y = \sqrt{x-2}$$

$$x \geq 2$$

■175 (2)

$$y = \frac{-3}{\sqrt{x+1}}$$

$$x > -1$$

■175 (3)

$$y = \sqrt{-x^2 + x + 6}$$

$$= \sqrt{-(x^2 - x - 6)}$$

$$= \sqrt{-(x-3)(x+2)}$$

$$-(x-3)(x+2) \geq 0$$

$$\therefore -2 \leq x \leq 3$$

■176 (1)

$$y = \sqrt{x-2} \quad (x \geq 2)$$

値域: $y \geq 0$

■176 (2)

$$y = \sqrt{x} - 2 \quad (x \geq 0)$$

値域: $y \geq -2$

■176 (3)

$$y = \sqrt{x-2} - 2 \quad (x \geq 2)$$

値域: $y \geq -2$

■176 (4)

$$y = \sqrt{x+1} + 2 \quad (x \geq -1)$$

値域: $y \geq 2$

■177 (1)

$$-y = x^3 + 3x^2$$

$$y = -x^3 - 3x^2$$

■177 (2)

$$y = (-x)^3 + 3(-x)^2$$

$$= -x^3 + 3x^2$$

■177 (3)

$$-y = (-x)^3 + 3(-x)^2$$

$$y = x^3 - 3x^2$$

■178

$$y = \sqrt{x-1} + 2$$

(1) x 軸対称

$$y = -\sqrt{x-1} - 2$$

(2) y 軸対称

$$y = \sqrt{-x-1} + 2$$

(3) 原点对称

$$y = -\sqrt{-x-1} - 2$$

■179 (1)

$$y = \frac{1}{x-2} \quad (x \neq 2, y \neq 0)$$

$$x-2 = \frac{1}{y}$$

$$x = \frac{1}{y} + 2$$

逆関数は $y = \frac{1}{x} + 2$

$(x \neq 0, y \neq 2)$

■179 (2)

$$y = x^2 - 3 \quad (x \geq 0, y \geq -3)$$

$$x^2 = y + 3$$

$$x = \sqrt{y + 3} \quad (\because x \geq 0)$$

$$\text{逆関数は } y = \sqrt{x + 3}$$

$$(x \geq -3, y \geq 0)$$

■179 (3)

$$y = -\sqrt{x} \quad (x \geq 0, y \leq 0)$$

$$y^2 = x$$

$$x = y^2$$

$$\text{逆関数は } y = x^2$$

$$(x \leq 0, y \geq 0)$$

■179 (4)

$$y = \sqrt{2 - x} - 1 \quad (x \leq 2, y \geq -1)$$

$$2 - x = (y + 1)^2$$

$$x = -(y + 1)^2 + 2$$

$$\text{逆関数は } y = -(x + 1)^2 + 2$$

$$(x \geq -1, y \leq 2)$$

CHECK

■180 (1)

$$\begin{aligned} f(-x) &= -2x \\ &= -f(x) \\ \therefore \text{奇関数} \end{aligned}$$

■180 (2)

$$\begin{aligned} f(-x) &= 3(-x)^2 - 1 \\ &= 3x^2 - 1 = f(x) \\ \therefore \text{偶関数} \end{aligned}$$

■180 (3)

$$\begin{aligned} f(-x) &= -x\{(-x)^2 - 1\} \\ &= -x(x^2 - 1) = -f(x) \\ \therefore \text{奇関数} \end{aligned}$$

■180 (4)

$$\begin{aligned} f(-x) &= \frac{(-x)^4 - 1}{(-x)^2} \\ &= \frac{x^4 - 1}{x^2} = f(x) \\ \therefore \text{偶関数} \end{aligned}$$

■180 (5)

$$\begin{aligned} f(-x) &= 3 \\ &= f(x) \\ \therefore \text{偶関数} \end{aligned}$$

■180 (6)

$$\begin{aligned} f(-x) &= 3 + (-x)^3 \\ &= 3 - x^3 \\ \therefore \text{いずれでもない} \end{aligned}$$

■181 (1)

$$\begin{aligned} y &= \frac{3x - 5}{x - 2} \\ &= 3 + \frac{1}{x - 2} \quad (x \neq 2) \\ \text{漸近線 } x &= 2, y = 3 \end{aligned}$$

■181 (2)

$$\begin{aligned} y &= -\sqrt{x+3} + 4 \quad (x \geq -3) \\ \text{値域: } y &\leq 4 \end{aligned}$$

■182

$$\begin{aligned} y + 3 &= \frac{2}{x - 1} \\ y &= \frac{2}{x - 1} - 3 \end{aligned}$$

■183

$$\begin{aligned} y - 2 &= \sqrt{-(x+3)} \\ y &= \sqrt{-x-3} + 2 \end{aligned}$$

■184 (1)

$$y = (x - 1)^3 - 2$$

■184 (2)

$$y = -(x + 1)^4 + 1$$

■184 (3)

$$\begin{aligned} y &= \frac{1}{x + 1} - 2 \\ \text{漸近線 } x &= -1, y = -2 \end{aligned}$$

■184 (4)

$$\begin{aligned} y &= \frac{2x + 3}{x + 2} \\ &= 2 - \frac{1}{x + 2} \\ \text{漸近線 } x &= -2, y = 2 \end{aligned}$$

■184 (5)

$$y = \sqrt{x+1} - 2 \quad (x \geq -1)$$

■184 (6)

$$\begin{aligned} y &= \sqrt{-x+1} + 1 \\ &= \sqrt{-(x-1)} + 1 \quad (x \leq 1) \end{aligned}$$

■185

$$\begin{aligned} y &= 3x^4 - 2x \\ &\rightarrow y = 3(-x)^4 - 2(-x) \\ y &= 3x^4 + 2x \end{aligned}$$

■186

$$\begin{aligned} y &= \sqrt{x+1} - 1 \\ \rightarrow -y &= \sqrt{-x+1} - 1 \\ y &= -\sqrt{-x+1} + 1 \end{aligned}$$

■187

$$\begin{aligned} y &= \frac{3}{x+2} \quad (x \neq -2, y \neq 0) \\ x+2 &= \frac{3}{y} \\ x &= \frac{3}{y} - 2 \\ \text{逆関数は } y &= \frac{3}{x} - 2 \\ &\quad (x \neq 0, y \neq -2) \end{aligned}$$

■188 (1)

$$\begin{aligned} f(x) &= (x-2)^2 \quad (x \leq 2, f(x) \geq 0) \\ \pm\sqrt{f(x)} &= x-2 \\ x &= 2 - \sqrt{f(x)} \quad (\because x \leq 2) \\ \text{よって } g(x) &= 2 - \sqrt{x} \quad (x \geq 0) \end{aligned}$$

■188 (2)

交点の x 座標は

$$(x-2)^2 = 2 - \sqrt{x} \text{ の解}$$

$$\sqrt{x} = 2 - (x-2)^2$$

$$(\text{LHS}) \geq 0 \text{ なので } (\text{RHS}) \geq 0$$

$$\therefore (x-2)^2 \leq 2$$

$$\text{すなわち } -\sqrt{2} \leq x-2 \leq \sqrt{2} \quad \dots \textcircled{1}$$

また $f(x), g(x)$ の定義域より

$$0 \leq x \leq 2 \quad \dots \textcircled{2}$$

このとき両辺 0 以上なので 2 乗して

$$x = \{2 - (x-2)^2\}^2$$

$$x = 4 - 4(x-2)^2 + (x-2)^4$$

$$(x-2)^4 - 4(x-2)^2 - (x-2) + 2 = 0$$

$$t = x-2 \text{ とすると}$$

$$t^4 - 4t^2 - t + 2 = 0$$

$$(\textcircled{1}, \textcircled{2} \text{ より } -\sqrt{2} \leq t \leq 0)$$

$$(t-2)(t^3 + 2t^2 - 1) = 0$$

$$(t-2)(t+1)(t^2 + t - 1) = 0$$

$$t = 2, -1, \frac{-1 \pm \sqrt{5}}{2}$$

$$-\sqrt{2} \leq t \leq 0 \text{ をみたすのは } t = -1$$

$$\text{このとき } x = 1$$

$$f(1) = (1-2)^2 = 1$$

$$\therefore \text{交点 } (1, 1)$$

STEP UP

■189 (1)

$$\begin{aligned} y &= \frac{3x-1}{2x-2} \\ &= \frac{3}{2} + \frac{2}{2x-2} \\ &= \frac{1}{x-1} + \frac{3}{2} \\ \text{漸近線 } x &= 1, y = \frac{3}{2} \end{aligned}$$

■189 (2)

$$\begin{aligned} y &= \frac{2x}{2x+3} \\ &= 1 - \frac{3}{2x+3} \\ &= -\frac{3}{2(x+\frac{3}{2})} + 1 \\ \text{漸近線 } x &= -\frac{3}{2}, y = 1 \end{aligned}$$

■190 (1)

$$y = \sqrt{-3x} \quad (x \leq 0)$$

■190 (2)

$$\begin{aligned} y &= \sqrt{-3x+6} \\ &= \sqrt{-3(x-2)} \quad (x \leq 2) \end{aligned}$$

■191

$$\begin{aligned} y &= \frac{4x-9}{2x-3} \\ &= 2 - \frac{3}{2x-3} \\ &\quad \left(-1 \leq x \leq 3, x \neq \frac{3}{2} \right) \\ \text{値域: } y &\leq 1, \frac{13}{5} \leq y \end{aligned}$$

■192

$$\begin{aligned} y &= \sqrt{-3x+a} \quad (-7 \leq x \leq b) \\ -b &\leq -x \leq 7 \\ -3b &\leq -3x \leq 21 \\ -3b+a &\leq -3x+a \leq 21+a \\ -3b+a &\leq y^2 \leq 21+a \\ 1 &\leq y^2 \leq 25 \text{ なので} \\ \begin{cases} -3b+a=1 \\ 21+a=25 \end{cases} &\text{ より } \begin{cases} a=4 \\ b=1 \end{cases} \\ \text{このとき } -3x+a &\geq 0 \text{ をみたす} \end{aligned}$$

■193

$$\begin{aligned} x &= \frac{2}{3}, y = 2 \text{ を漸近線とする分数関数は} \\ y &= \frac{k}{x-\frac{2}{3}} + 2 \quad (k: \text{定数}) \\ (1, 1) &\text{ を通るので} \\ 1 &= \frac{k}{\frac{1}{3}} + 2 \\ k &= -\frac{1}{3} \\ y &= -\frac{1}{3x-2} + 2 \\ &= \frac{6x-5}{3x-2} \\ \text{これが } y &= \frac{ax+b}{3x+c} \text{ に一致するので} \\ a &= 6, b = -5, c = -2 \end{aligned}$$

■194 (1)

$$\begin{aligned} y &= \frac{a}{x+2} + b \text{ とおける } (a, b: \text{定数}) \\ (2, 4), (-1, 7) &\text{ を通るので} \\ \begin{cases} 4 = \frac{1}{4}a + b \\ 7 = a + b \end{cases} &\text{ より } \begin{cases} a = 4 \\ b = 3 \end{cases} \\ y &= \frac{4}{x+2} + 3 \end{aligned}$$

■194 (2)

$$y = \frac{b}{x-a} + 2 \text{ とおける } (a, b: \text{定数})$$

(4, 6), (2, -2) を通る

$$\begin{cases} 6 = \frac{b}{4-a} + 2 \\ -2 = \frac{b}{2-a} + 2 \end{cases}$$

$$\begin{cases} 4(4-a) = b \\ -4(2-a) = b \end{cases}$$

$$16 - 4a = -8 + 4a$$

$$a = 3$$

$$b = 4$$

$$y = \frac{4}{x-3} + 2$$

■194 (3)

$$y - a = \frac{3}{x-b} \text{ とおける } (a, b: \text{定数})$$

(-2, 8), (-4, 10) を通るので

$$\begin{cases} 8 - a = \frac{3}{-2-b} \\ 10 - a = \frac{3}{-4-b} \end{cases}$$

$$2 = -\frac{3}{4+b} + \frac{3}{2+b}$$

$$2(b+2)(b+4) = -3(b+2) + 3(b+4)$$

$$2(b^2 + 6b + 8) = 6$$

$$b^2 + 6b + 5 = 0$$

$$(b+1)(b+5) = 0$$

$$b = -1, -5$$

$$a = 8 + \frac{3}{b+2} \text{ なので}$$

$$(a, b) = (11, -1), (7, -5)$$

$$y = \frac{3}{x+1} + 11, \quad y = \frac{3}{x+5} + 7$$

■195

$$y = \frac{3}{3x+k} + 2(k+1) \quad \dots \textcircled{1}$$

$$\left(= \frac{1}{x + \frac{k}{3}} + 2(k+1) \right)$$

$$y - 2(k+1) = \frac{3}{3x+k}$$

$y \neq 2(k+1)$ なので逆数をとって

$$\frac{1}{y - 2(k+1)} = \frac{3x+k}{3}$$

$$x = \frac{1}{y - 2(k+1)} - \frac{k}{3}$$

$\therefore$ 逆関数は

$$y = \frac{1}{x - 2(k+1)} - \frac{k}{3}$$

これが ① に一致するので

$$\begin{cases} \frac{k}{3} = -2(k+1) \\ 2(k+1) = -\frac{k}{3} \end{cases}$$

$$k = -6k - 6$$

$$k = -\frac{6}{7}$$

■196

$$g(5) = 1 \Leftrightarrow f(1) = 5$$

$$g(2) = 4 \Leftrightarrow f(4) = 2$$

$$\therefore \begin{cases} 5 = \frac{a+2}{2-b} \\ 2 = \frac{4a+2}{8-b} \end{cases}$$

$$\begin{cases} 10 - 5b = a + 2 \\ 8 - b = 2a + 1 \end{cases} \quad \text{より} \quad \begin{cases} a = 3 \\ b = 1 \end{cases}$$

■197

$$y = 2\sqrt{x+1} \text{ と } y = \frac{1}{3}x + k$$

が接するときを考える

$$2\sqrt{x+1} = \frac{1}{3}x + k$$

両辺 2 乗して

$$36(x+1) = (x+3k)^2$$

$$x^2 + 6(k-6)x + 9(k^2-4) = 0$$

判別式を D とすると

$$\begin{aligned} \frac{D}{4} &= 9(k-6)^2 - 9(k^2-4) \\ &= 9(k^2 - 12k + 36) - 9(k^2 - 4) \\ &= 9(-12k + 40) \end{aligned}$$

$$\text{重解をもつとき } -12k + 40 = 0$$

$$k = \frac{10}{3}$$

このとき交点の x 座標は

$$x^2 - 16x + 64 = 0$$

$$(x-8)^2 = 0$$

$$x = 8, y = 6$$

また $(-1, 0)$ を $y = \frac{1}{3}x + k$ が通るとき

$$0 = -\frac{1}{3} + k$$

$$k = \frac{1}{3}$$

$\therefore$ グラフより

$$k < \frac{1}{3} \text{ のとき } 1 \text{ 個}$$

$$\frac{1}{3} \leq k < \frac{10}{3} \text{ のとき } 2 \text{ 個}$$

$$k = \frac{10}{3} \text{ のとき } 1 \text{ 個}$$

$$\frac{10}{3} < k \text{ のとき } 0 \text{ 個}$$

■198

交点の x 座標は

$$\frac{1}{x} = -\frac{1}{4}x + k \text{ の解 } (x \neq 0)$$

$$4 = -x^2 + 4kx$$

$$x^2 - 4kx + 4 = 0 \quad \dots \textcircled{1}$$

判別式を D とすると

$$\begin{aligned} \frac{D}{4} &= 4k^2 - 4 \\ &= 4(k+1)(k-1) < 0 \\ &-1 < k < 1 \end{aligned}$$

また $x = 0$ は $\textcircled{1}$ の解でない

$\therefore$ 共有点をもたないとき $-1 < k < 1$

PLUS

■199 (1)

$$\begin{aligned} y &= |x^2 - 1| \\ &= \begin{cases} x^2 - 1 & (x \leq -1, 1 \leq x) \\ -x^2 + 1 & (-1 < x < 1) \end{cases} \end{aligned}$$

■199 (2)

$$\begin{aligned} y &= x^2 - 2|x| + 1 \\ &= \begin{cases} x^2 + 2x + 1 & (x < 0) \\ x^2 - 2x + 1 & (x \geq 0) \end{cases} \\ &= \begin{cases} (x+1)^2 & (x < 0) \\ (x-1)^2 & (x \geq 0) \end{cases} \end{aligned}$$

■200 (1)

$y = x$ と $y = -\frac{1}{x}$ のグラフをたしたもの
漸近線 $x = 0, y = x$

■200 (2)

$y = x$ と $y = \frac{1}{x}$ のグラフをたしたもの
漸近線 $x = 0, y = x$